# MORGAN R. PFAFF

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## EDUCATION

**University of California (UC) Berkeley and UC San Francisco (UCSF)** **Berkeley, CA**
Doctor of Philosophy, UC Berkeley–UCSF Joint Graduate Program in Bioengineering October 2026

- Dissertation: "Semi-synthetic Hyaluronic Acid Hydrogels to Direct Fibro-Adipogenic Progenitor Fate in Injured Rotator Cuff Muscle"; Advisor: Dr. Kevin Healy
- Major: Biomaterials Engineering & Tissue Regeneration; Minor: Cell, Tissue, & Developmental Biology
- GPA 4.0/4.0; National Science Foundation Graduate Research Fellow

**Northeastern University** **Boston, MA**
Bachelor of Science in Biochemistry May 2021

- GPA 4.0/4.0; Honors Program; Honors in the Discipline
- Engineering & entrepreneurship study abroad program, Oxford, United Kingdom (Summer 2019)

## RESEARCH EXPERIENCE

**Kevin Healy Lab, UC Berkeley** **Berkeley, CA**
Doctoral Researcher August 2021 – October 2026

- Engineered viscoelastic hyaluronic acid (HyA) hydrogels for rotator cuff muscle regeneration; characterized rheology, injectability, and chemical composition of different formulations; and developed high-throughput silicone (PDMS) culture arrays that enabled micro-scale 3D testing of hydrogels with precious patient-derived cells
- Used single-cell transcriptomics of patient-derived fibro-adipogenic progenitors (FAPs) to identify how these cells sense and remodel their surroundings, and applied those findings to develop biologically informed hydrogels
- Demonstrated that the hydrogel shifts FAPs toward a pro-regenerative beige phenotype and studied its effects on muscle stem cell fusion in 3D cultures and cocultures
- Revived and updated historical lab methods, improved hydrogel fabrication and characterization protocols, and helped coordinate reagent synthesis, inventory, and quality control
- Prepared hydrogels and helped plan and interpret collaborators' *in vivo* studies at UCSF and the University of Virginia, including mouse studies in which the hydrogel reduced fibrosis and fat infiltration and improved function
- Published 2 peer-reviewed papers (1 first-author) and 3 in preparation, gave 5 presentations at 4 national and international conferences, and contributed data and writing to 8+ grant proposals, including an NIH R01

**Randall Erb Lab, Northeastern University** **Boston, MA**
Undergraduate Researcher September 2016 – July 2021

- Developed dual-network hydrogels reinforced with magnetically responsive hydroxyapatite fibers for bone grafts, characterizing their mechanics and microstructure (dynamic mechanical analysis, scanning electron microscopy)
- Built COMSOL finite element simulations to explore fiber rotation and stress distribution as potential toughening mechanisms in composite materials
- Fabricated freeze-cast collagen conduits with aligned, double-layered porosity for peripheral nerve repair in collaboration with Dartmouth College and Harvard Medical School

**Marc Hansen Lab, University of Connecticut Health Center** **Farmington, CT**
Summer Research Fellow June – August 2017; June – August 2018

- Piloted synthetic-lethal inhibitor and siRNA combinations in osteosarcoma and normal osteoblast cells to find druggable vulnerabilities in growth factor receptor signaling, providing preliminary data for NIH grant applications
- Isolated osteoblast and osteoclast populations from histological sections using laser capture microdissection to investigate cell behavior and genetics in Paget's disease of bone

## TRANSLATIONAL, REGULATORY, & INDUSTRY EXPERIENCE

**MuscleMatrix** **Berkeley, CA**

Scientific Consultant August 2026 – Present

- Advise an early-stage startup, spun out in part from my doctoral research and co-invented technology, that is developing an injectable, cell-free hyaluronic acid scaffold for muscle regeneration after rotator cuff injury
- Shape FDA regulatory strategy for a medical device or combination product, and help write grant applications
- Build company operations from the ground up, including regulatory and environmental health and safety compliance, lab buildout, equipment procurement, manufacturer and supplier sourcing, and staff training
- Characterize how drug loading affects injectability and mechanics to inform a planned drug-eluting version

**Lake Street Labs** **Remote**

Scientific Consultant July 2026 – August 2026

- Conducted a scientific and regulatory feasibility review of a novel product for a startup, synthesizing clinical and preclinical literature to guide product development
- Proposed candidate formulations, evaluating chemical compatibility, stability, storage, packaging, and shelf-life testing needs
- Assessed FDA pathways, product classification, and permissible claims, and delivered recommendations to company leadership through discussion meetings and a final written report

**Vertex Pharmaceuticals** **Boston, MA**

Exploratory Research Co-op January 2020 – June 2020

- Developed and optimized a reporter assay to measure transcription factor activity in potentially druggable sickle cell disease (SCD) pathways, giving the team a faster, more accurate readout of compound effectiveness, and built the stable cell lines it required by molecular cloning
- Independently planned, executed, and analyzed experiments, and presented results and journal club articles to the SCD exploratory research team

**Editas Medicine** **Cambridge, MA**

Screening Group Co-op January 2018 – June 2018

- Screened sgRNAs to identify the most effective candidates for the company's pipeline diseases, introducing CRISPR-mediated edits via nucleofection, transfection, and viral transduction
- Programmed and optimized high-throughput screening assays on automated liquid handlers, enabling more complex, higher-throughput sgRNA screening; analyzed data in Prism, Tableau, and FlowJo and presented findings to the screening group and other departments

## TEACHING & MENTORING

**University of California, Berkeley** **Berkeley, CA**

BIOE/MSE C118/C208: Biological Performance of Materials August 2022 – Present

- Graduate Student Instructor (Fall 2022; 45 students), Reader (Fall 2026; 30 students), and Guest Lecturer (2023–2026) for Dr. Kevin Healy's course
- Delivered 4 full-length lectures (2 on hydrogels, 1 on polymer viscoelasticity, and 1 introduction to polymers)
- Led weekly discussion sections and office hours, and evaluated ~10 team design projects, including 15-page proposals for biomaterial solutions to unmet clinical needs and pitch, 10-, and 20-minute presentations
- Organize student design teams and grade homework, exams, written reports, and presentations

BIOE 252: Clinical Need-Based Therapy Solutions August 2026 – Present

- Reader for Dr. Syed Hossainy's course (~100 students): guide teams developing therapy concepts for unmet clinical needs, grade coursework, and coordinate guest lecturers from around the country

Research Mentor, Kevin Healy Lab August 2022 – October 2026

- Undergraduate researcher (2023–2024): integrated hydrogels into a microphysiological system to study FAP migration in a physiologically relevant model; now a PhD student at Johns Hopkins University

- Master of Translational Medicine capstone team (4 students, 2024–2025): characterized hydrogel rheology to assess injectability and conducted a market analysis for rotator cuff injury indications
- Visiting PhD student from Spain (2024): fabricated carbon nanotube–HyA hydrogel composites, tested their conductivity, and assessed the beating behavior of iPSC-derived cardiomyocytes encapsulated in them
- 3 first-year PhD rotation students (2022–2026): developed a degradation-based method to isolate cells from HyA hydrogels, optimized long-duration rheology testing, and built machine learning–based screens for FAP fate
- Trained dozens of additional students and lab members on equipment and methods

## PUBLICATIONS & PRESENTATIONS

**Publications**

1. Wang, V., Zhu, Y., Spaulding, K., **Pfaff, M.**, Neiman, G., Takaba, K., Chen, N., Parekh, K., Hasaballa, A., Haraldsson, H., Baker, A., Saloner, D., Wallace, A., Ziats, N., Healy, K., Ratcliffe, M. NIPAAm-co-PEG4000 thermoresponsive hydrogel with ROS scavenging properties improves in vivo border zone contractility and reduces myocardial remodeling in sheep after MI. *Acta Biomaterialia*, 2026, 211, 48–60. doi.org/10.1016/j.actbio.2025.06.046
2. **Pfaff, M.**, Wague, A., Davies, M., Killaars, A., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Viscoelastic HyA hydrogel promotes recovery of muscle quality and vascularization in a murine model of delayed rotator cuff repair. *Advanced Healthcare Materials*, 2025, 14, 2403962. doi.org/10.1002/adhm.202403962

**Manuscripts in Preparation**

1. **Pfaff, M.**, Chen, N., Jilani, H., Garcia, S., Lau, J., Wague, A., Hohman, O., Liu, X., Feeley, B., Healy, K. Transcriptome-guided design of hyaluronic acid hydrogels to direct human fibro-adipogenic progenitor fate. In preparation.
2. **Pfaff, M.**, Sagar, R., Andrade, B., Hines Ramirez, D., Pal, S., Goswami, I., Messersmith, P., Healy, K. Bioinspired adhesive for reversibly bonded microphysiological systems. In preparation for *Lab on a Chip*.
3. Wague, A., **Pfaff, M.**, Fusagawa, H., Lau, J., Sharma, S., Youn, A., Healy, K., Liu, X., Feeley, B. Hydrogel-delivered beige adipose precursors enhance muscle quality and functional recovery in a delayed rotator cuff repair mouse model. In preparation for *Cell Biomaterials*.

**Oral Presentations**

1. **Pfaff, M.**, Wague, A., Davies, M., Killaars, A., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Viscoelastic HyA hydrogel promotes recovery of muscle quality and vascularization in a murine model of delayed rotator cuff repair. Presented at: Biological Scaffolds for Regenerative Medicine (BSRM); May 2025; Napa, CA.
2. **Pfaff, M.**, Killaars, A., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Presented at: Biomedical Engineering Society (BMES) Annual Meeting; October 2023; Seattle, WA.
3. **Pfaff, M.**, Killaars, A., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Presented at: Society for Biomaterials (SFB) Annual Meeting & Exposition, "Biomaterials for Regenerative Engineering"; April 2023; San Diego, CA.
4. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Presented at: Materials Research Society 2020 Spring/Fall Meeting and Exhibit; November 2020; Virtual.
5. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Presented at: Northeastern Regional Student Conference of the Society for Experimental Mechanics; June 2020; Virtual.

**Poster Presentations**

1. **Pfaff, M.**, Killaars, A., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Presented at: Orthopaedic Research Society (ORS) 50th International Musculoskeletal Biology Workshop; July 2023; Midway, UT.

2. **Pfaff, M.**, Killaars, A., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B., Healy, K. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Presented at: Society for Biomaterials (SFB) Annual Meeting & Exposition; April 2023; San Diego, CA.
3. **Pfaff, M.**, Erb, R. Understanding rod rotation in fiber reinforced hydrogels: a potential toughening mechanism. Presented at: Research, Innovation, and Scholarship Expo; April 2021; Virtual.
4. **Pfaff, M.**, Alander, C., Hansen, M. Synthetic lethal approaches to treating osteosarcoma. Presented at: University of Connecticut Health Center Summer Research Fellowship Program; July 2017; Farmington, CT.
5. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Presented at: Research, Innovation, and Scholarship Expo; April 2017; Boston, MA.
6. **Pfaff, M.**, Faust, J., Erb, R. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Presented at: National Conference on Undergraduate Research; April 2017; Memphis, TN.

## FELLOWSHIPS & AWARDS

National Science Foundation Graduate Research Fellowship ($138,000) 2021
Research, Innovation, and Scholarship Expo (RISE) Excellence in Research Award ($2,000) 2021
PEAK Summit Award, undergraduate research grant ($3,000) 2020
PEAK "Shout It Out!" Award, conference travel grant 2020
RISE Innovation Alley Selection (poster selected for highest impact and commercial potential) 2017

## LEADERSHIP & SERVICE

**Food Not Bombs East Bay** **September 2024 – Present**

- Lead a weekly shift, picking up meals prepared by volunteer cooks and serving 25–40 free vegetarian meals to unhoused and low-income community members
- Transformed an unstaffed curbside drop-off into an in-person meal service, creating a more personal experience and building ongoing relationships with community members

**Sigma Xi, The Scientific Research Honor Society** **2020 – Present**

UC Berkeley Chapter, Vice President 2023 – Present
UC Berkeley Chapter, Founder & Chair, Committee for Student Advancement 2021 – Present

- Grew student involvement in the committee from 2 to ~35 undergraduate and graduate students; lead a 3–6-person board that meets weekly to biweekly
- Mentored 17 students through Sigma Xi membership nominations and oversaw a board-run voluntary mentorship program pairing undergraduate pre-members with associate members
- Organized 20+ chapter events, including a 5-part student and postdoc seminar series, 11 community-building socials, 4 new-member info sessions, and an induction ceremony; hosted the Editor-in-Chief of *American Scientist*, Dr. Fenella Saunders, for a Faculty Club lecture and dinner
- Planned and ran the Sigma Xi–ATP Bio Research Symposium & Poster Session on my own, from securing a funding partner to day-of logistics, drawing ~25 presenters (including first-time presenters and students from nearby universities and community colleges), a keynote speaker, and ~150 attendees
- Co-organize the semi-annual Grants in Aid of Research program, awarding up to $1,250 to undergraduate, master's, and PhD students across STEM, and lead chapter recruitment through campus groups and departments

Northeastern University Chapter, Diversity & Inclusion Chair 2020 – 2021

- Expanded outreach to first-generation, culture-specific, and other underrepresented student organizations, and hosted a campus-wide diversity, equity, and inclusion panel

**BioEngineering High School Competition (BioEHSC)** **April 2022 – April 2026**

- Academic Judge: evaluated 10–15 high school team pitches each spring in a full-day bioscience pitch competition, assessing depth of knowledge through Q&A and providing feedback

**Northeastern University Service & Leadership** **September 2016 – June 2021**

- Triathlon Team President/Captain (2019–2020), Chief Financial Officer (2018–2019), and Chief Procurement Officer (2017–2018); Club Sports Council liaison between teams and administration (2019–2021)
- Resident Assistant (2017–2019): live-in mentor for ~45 first-year women in STEM each year, designing programs such as Q&A panels with graduate students and professors and faculty mentorship events
- Volunteer swim mentor for a Boston Children's Hospital patient with a disability through Adventure Challenge (2018–2019), and AVID college-readiness tutor at Fenway High School (2018)

## CLINICAL EXPERIENCE

**Tufts Medical Center, Oncology and Hematology** **Boston, MA**

Clinical Care Technician January 2019 – December 2019

- Provided direct care for 6–10 oncology and hematology patients per shift, monitoring vital signs, escalating abnormal findings to nursing staff, and supporting clinical teams during procedures and emergencies

**Fallon Ambulance Service** **Quincy, MA**

Emergency Medical Technician (EMT-Basic) September 2018 – April 2019

- Assessed, treated, and transported ill and injured patients, providing immediate medical and psychological care en route to hospitals and specialty facilities

**Dental Office of Dr. Andrew J. Ponichtera, DMD, MS** **Simsbury, CT**

Dental Assistant June 2016 – August 2016

- Supported chairside procedures, including fillings, root canals, crowns, implants, bridges, and dentures

## SKILLS

**Biomaterials Synthesis & Mechanical Testing**

- Hydrogel synthesis and chemical modification, drug loading and release, rheology, injectability testing, mechanical testing (dynamic mechanical analysis, tensile, burst, and lap shear), freeze casting, lyophilization

**Materials Characterization & Microfabrication**

- NMR spectroscopy, X-ray photoelectron spectroscopy (XPS), scanning electron microscopy (SEM), microfabrication and photolithography

**Cell & Molecular Biology**

- Primary and 3D cell culture, immunocytochemistry, confocal and fluorescence microscopy, histology, flow cytometry, qPCR/ddPCR, Western blot, molecular cloning, CRISPR-Cas9 gene editing (transfection, nucleofection, viral transduction), automated liquid handling

**Data Analysis & Computation**

- R and Python (incl. single-cell RNA-seq), GraphPad Prism, MATLAB, COMSOL (finite element analysis)

**Regulatory & Science Communication**

- FDA pathway and product classification assessment, regulatory feasibility review, scientific literature synthesis, grant and technical report writing